



OPM Exam

Exercise 01 :

- 1 - Create a vector 'V' ranging from 1 to 11 with a step of 2.
- 2 - Replace the 2nd and 5th elements of 'V' with zero.
- 3 - Find the values of the following elements:
A = 2*ones (1,3); B = 2*eye (4); C = [V (3) A(1) (B(2,2)-1) (V(4)+1); B(2,:)] ; D = B.^2;
- 4- Correct the errors in the following MATLAB instructions:
D#16 = complex[1;2]; plot('x,f') ; subplot("4,3,2,1"); x = colon(1;0.5;3); m.2 = moy ([1:8]);

Exercise 02 :

- 1- Solve the following system using MATLAB:

$$\begin{cases} -2x + 6z - 2k = 3 \\ 3y - z + k = -2 \\ x + 5z = 0 \\ 3x + 2y - 2k = 6 \end{cases}$$

- 2- Translate the following mathematical expression into MATLAB syntax:

$$z \leftarrow \frac{|3n^3-5|}{\sqrt{5n^3+2\ln(3n)}} e^{2-n}; y \leftarrow 2 \cos x + x^2 \arctan\left(\frac{1.5}{2x}\right)$$

Exercise 03:

- 1- Write a MATLAB program to solve the following problem:

$$\begin{aligned} y &= -2x + 5 & x < 0 \\ y &= 3x^2 & x > 0 \\ y &= 10 & x = 0 \end{aligned}$$

- 2- Provide MATLAB commands to compute the following sum (without using loops)

$$S = \pi + \frac{\pi}{4} + \frac{\pi}{9} + \frac{\pi}{16}$$

Exercise 04:

We consider the following polynomial: $P(x) = x^3 - 6x^2 - 72x + 27$

- Determine the roots of the polynomial P(x).
- Evaluate the polynomial at 'x = 3'.
- Compute the derivative and the integral of 'P(x)'.
- Plot the polynomial 'P(x)', with x = [-2 ,6].
- Overlay on the same graph the function $g(x) = \sqrt{3}xe^{2x+1}$.
- Add a title, legend, grid, and axis labels to the graph.

Good Luck
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